

# Lesson 0 – Introduction

## 0.1 DESCRIPTION

In this lesson, as well as in further lessons, different icons and font styles guide you and help you get the necessary information in order to complete these lessons.



Important general information



Information on the current chapter



Workflow (what you should do)

*Italic*

*Italic* letters show you the *command* name and *default* settings

**Bold**

**Bold** letters show you the **active** command name

SMALL CAPS

Mouse control movements

1.

General numeration



Numeration for program work steps

Text

Input text to type in

## 0.2 INSTALLING ArcGIS Pro (ON YOUR PRIVATE COMPUTER)

We will work with ArcGIS Pro this semester. You can either use the public computers in the CIP-Pools (computer labs) or your private computer to run it.

If you are using a public computer, ArcGIS Pro is already installed, so you can proceed directly to sections 0.3 and 0.4.

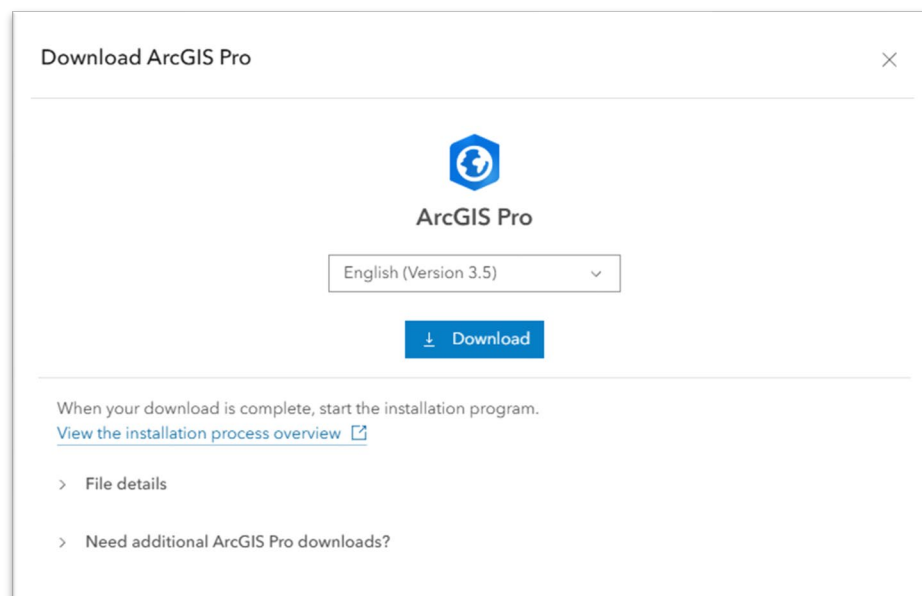
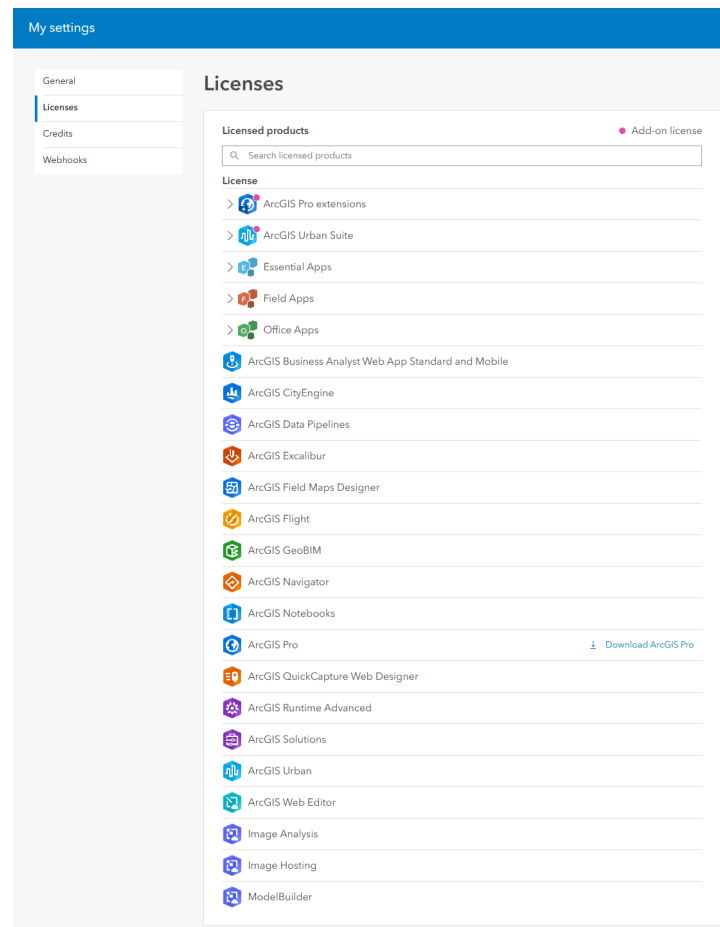
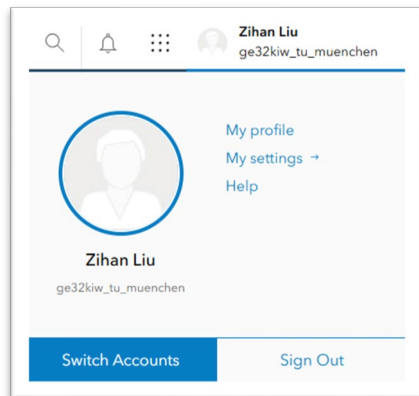
If you are using a private computer, please check out the following installation guide. It is worth noting that ArcGIS Pro can only run in a Windows environment.



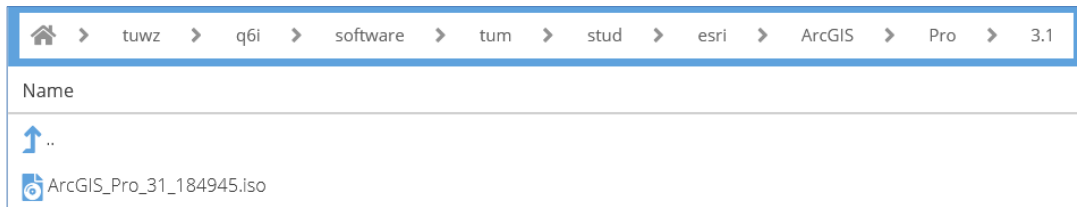
Activate your student license, download, install, and sign in to ArcGIS Pro.

1. Register at [ArcGIS Online](https://tu-muenchen.maps.arcgis.com) (<https://tu-muenchen.maps.arcgis.com>) to activate your student license. Click the button *TUM* and log in with your TUM credentials.

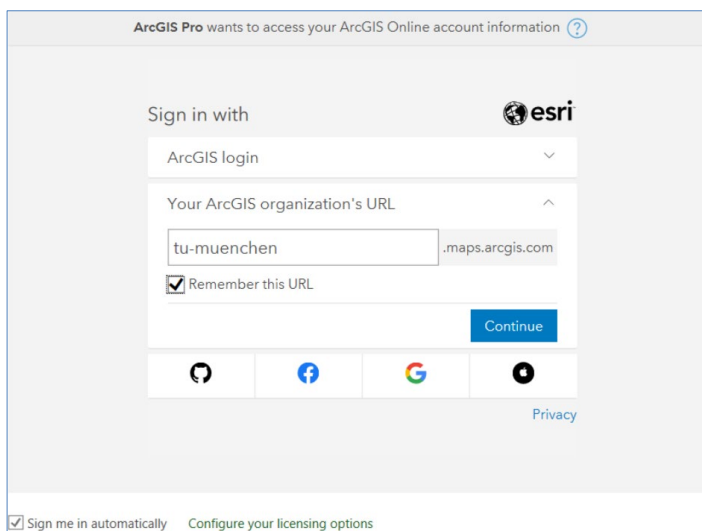
2. Go to *My Settings*, under *Licenses*, find *ArcGIS Pro*, then click on *Download ArcGIS Pro*, choose the *English* version, *Download* it, and install it.



- Alternatively, you can also download the installation file (\*.iso) to install ArcGIS Pro at <https://webdisk.ads.mwn.de/?folder=tuwz/q6i/software/tum/stud/esri> (under ArcGIS > Pro, anything from version 3.0 upwards is fine).



- After installing it, you can run ArcGIS Pro. Open the program and sign in using the URL [tu-muenchen.maps.arcgis.com](https://tu-muenchen.maps.arcgis.com) and your TUM credentials. Then, you can move to 0.5.



## 0.3 COMPUTER LOGIN TO TUM PC

You can log in with your TUM credentials on all computers in the Carto Lab (0712) as well as the other faculty CIP-Pools (3209, 3238, N0199).

- ❗ **Username:** your TUMonline username  
**Password:** your TUMonline password

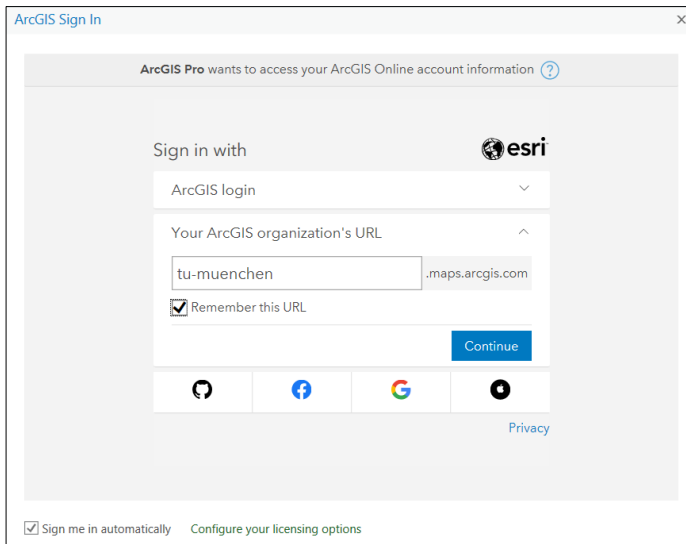
## 0.4 STARTING ArcGIS Pro

**W** Activate your student license, and sign in to ArcGIS Pro.

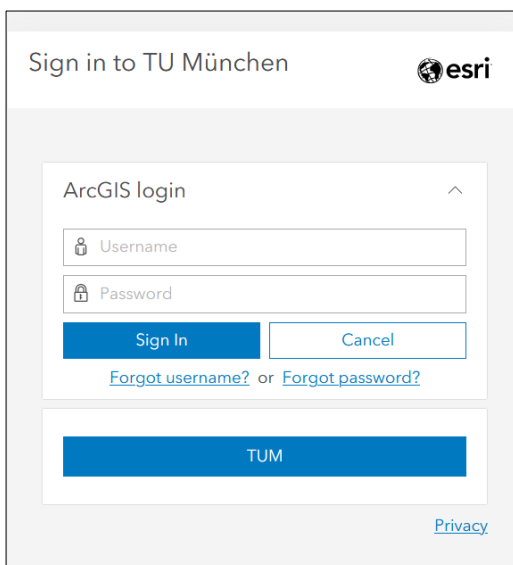
- If you start ArcGIS Pro for the first time at TUM, you will have to register at ArcGIS Online (<https://tu-muenchen.maps.arcgis.com>) to activate your student license.

Click the button **TUM** and log in with your TUM credentials. If you have already completed this step, you can proceed to the next step.

2. Run ArcGIS Pro and *Sign in* with the URL **tu-muenchen.maps.arcgis.com**



3. ArcGIS will open a browser tab for you to sign in. **CLICK** the blue TUM button and log in with your TUM credentials



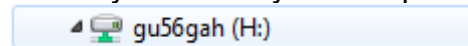
(Confirm occurring script error notices, they should be of no harm.)

## 0.5 DATA

You will find your data set on the course website (<https://services.lfk.lrg.tum.de/Geo-Information/exercises/E0.html#data>). Download the *Data Students.gdb* file to your local drive and unzip the file. This geodatabase features all the data you need for the first lessons. Additional data for forthcoming lessons will also be given in the coming weeks.

## 0.6 SAVING YOUR DATA ON THE TUM PC

If you are using a public computer, save all of your work on your personal webdisk or on a memory stick only. Your personal webdisk is the drive with your TUMonline name.



You can access your webdisk from a browser, too. <https://webdisk.ads.mwn.de/>

It is advisable to save all geodata files (\*.gdb, \*.shp, \*.dbf, etc.) and project files (\*.aprx) in the same folder. The project file does not feature any geodata!

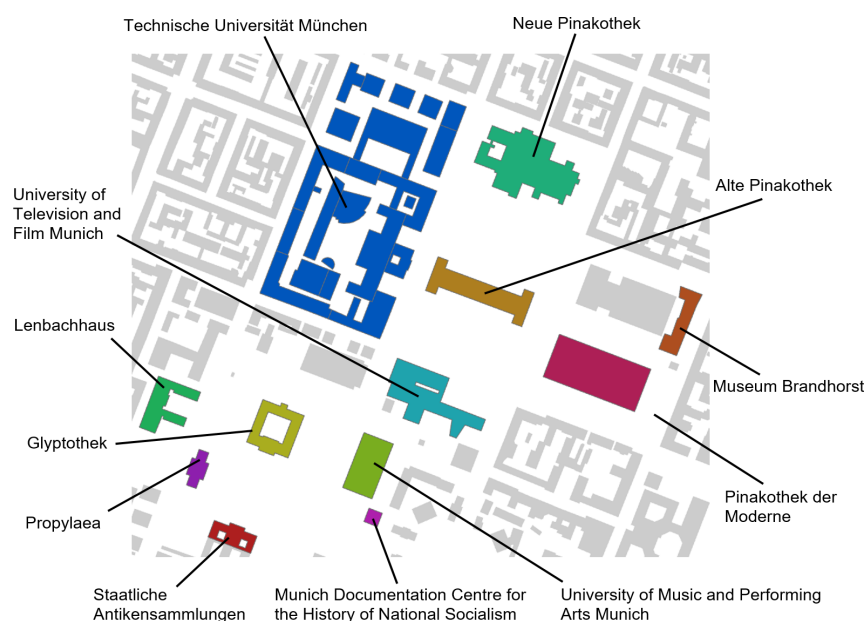
It's essential to save your work frequently; otherwise, if the program crashes, hours of work can be lost.

❗ **Save often and frequently under changing file names**

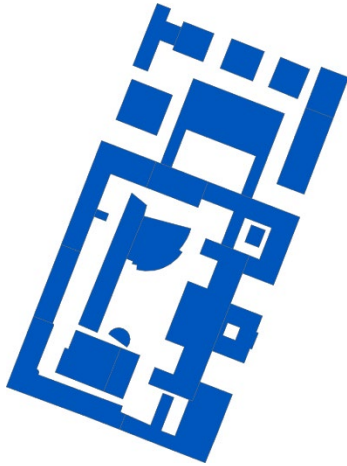
## 0.7 DIRECTORY AND PATH NAME

Never create directory names that contain **spaces**, **dots**, or other **special (non-alphanumeric) characters**, anywhere in the path name for the directory. This can lead to problems in ArcGIS Pro. Instead of spaces, use underscores ("Traffic\_Road\_Layer") or Camel styles ("TrafficRoadLayer").

## 0.8 STUDY AREA



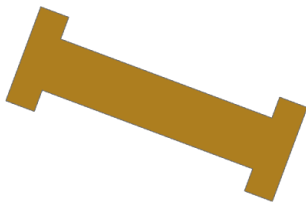
In the next few lessons, we will work with a data set located around the Technical University of Munich, namely [Kunstareal München](#) (art area of Munich). The figure above shows you the building outlines from the study area. The colored buildings are the most famous ones.



**Technische Universität München  
(Technical University of Munich, TUM)**

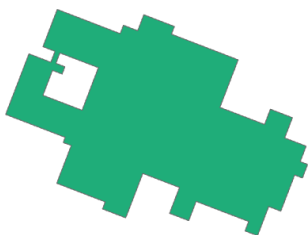
You know this one.

In 1868, King Ludwig II founded a “Polytechnic School,” which was later reorganized and renamed “Technical University” (Technische Hochschule) in 1877. It was later officially renamed the Technical University of Munich in 1970.



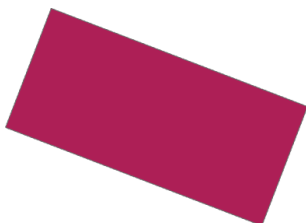
**Alte Pinakothek  
(Old Pinakothek)**

On display are more than 800 paintings, including numerous well-known masterpieces of 14th- to 18th-century European painting.



**Neue Pinakothek  
(New Pinakothek)**

Offers an overview of European art from classicism to Art Nouveau.



**Pinakothek der Moderne  
(Pinakothek of Modern Art)**

It is a Modern Art Museum uniquely housing four independent collections under one roof: art, architecture, graphics (prints and drawings), and design. It holds one of the world’s greatest collections of 20th- and 21st-century modern art.



### **Museum Brandhorst**

It displays about 200 exhibits from the collection of modern and contemporary art amassed by Udo and Anette Brandhorst, the latter a Henkel family heir. The full collection comprises over 1,200 works.



### **Glyptothek**

The Glyptothek was commissioned by King Ludwig I of Bavaria to house his collection of Greek and Roman sculptures.



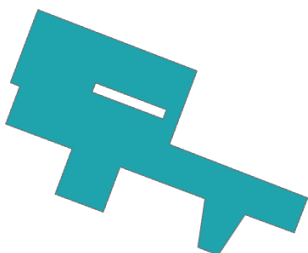
### **Propyläen**

This magnificent gate was completed in 1862 as a memorial to the accession to the throne of Otto of Greece, the son of King Ludwig I of Bavaria.



### **Lenbachhaus**

This gallery is particularly renowned for its extensive collection of works by artists associated with The Blue Rider (Der Blaue Reiter), a group central to Expressionism, alongside a diverse range of works by Munich painters and contemporary artists, including those of the New Objectivity movement.



### **Hochschule für Fernsehen und Film München (University of Television and Film Munich, HFF Munich)**

The University of Television and Film Munich is a publicly funded film school. It is one of Germany's most reputable film schools, with about 350 students enrolled.



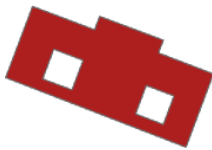
**NS-Dokumentationszentrum München  
(Munich Documentation Centre for the History of National  
Socialism)**

The Munich Documentation Centre for the History of National Socialism is a museum that focuses on the history and consequences of the Nazi regime, as well as the role of Munich as the *Hauptstadt der Bewegung* (capital of the movement).



**Hochschule für Musik und Theater München  
(University of Music and Performing Arts Munich)**

The University of Music and Performing Arts Munich is one of the most respected traditional vocational universities in Germany, specializing in music and the performing arts.



**Staatliche Antikensammlungen  
(State Collections of Antiquities)**

It is a museum holding Bavaria's collections of antiquities from Greece, Etruria, and Rome.